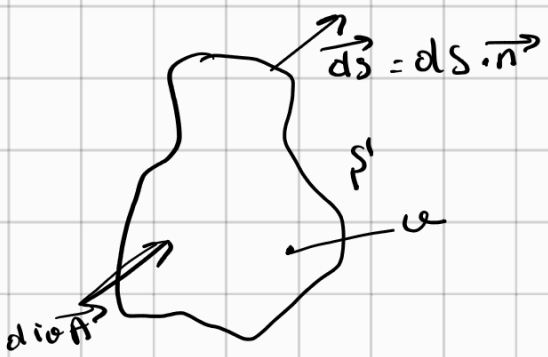
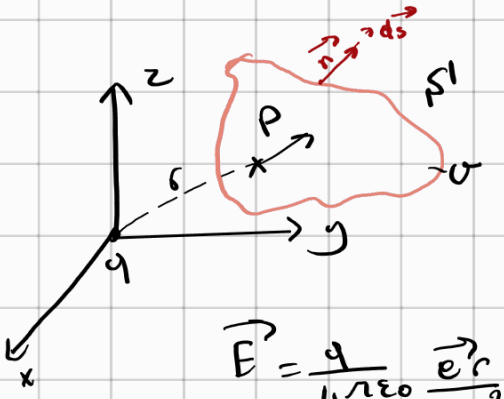
Gauss-Ostrogradski Teo:



$$\int_v \operatorname{div} \vec{A} = \int_{S'} \vec{A} \cdot ds$$



q v bölgesinin dışında

$$\vec{E} = \frac{q}{4\pi\varepsilon_0} \frac{\vec{e_r}}{r^2} = \frac{q \cdot \vec{r}}{4\pi\varepsilon_0 r^3}$$

$$\vec{D} = \varepsilon_0 \vec{E} = \frac{q}{4\pi} \frac{\vec{r}}{r^3}$$

Deplasman
(yer değiştirme vektörü)

$$\int_{S'} \vec{D} \cdot ds = \int_v \operatorname{div} \vec{D} \cdot dv = \int_v \operatorname{div} \left(\frac{q}{4\pi} \frac{\vec{r}}{r^3} \right) dv$$

$$= \frac{q}{4\pi} \int_v -\operatorname{div} \left(-\frac{\vec{r}}{r^3} \right) dv$$

$$= -\frac{q}{4\pi} \int_v \underbrace{\operatorname{div} \operatorname{grad} \left(\frac{1}{r} \right)}_{\Delta(1/r)} dv$$

$\frac{\partial}{\partial x}\vec{e}_x + \frac{\partial}{\partial y}\vec{e}_y + \frac{\partial}{\partial z}\vec{e}_z$

$f(x,y,z) \Rightarrow \operatorname{grad} f = \vec{\nabla} \cdot f$

$$= \frac{\partial f}{\partial x}\vec{e}_x + \frac{\partial f}{\partial y}\vec{e}_y + \frac{\partial f}{\partial z}\vec{e}_z$$

$\text{divgrad}\, f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} = \Delta f$ laplacian

$r = \sqrt{x^2+y^2+z^2} \Rightarrow \frac{1}{r} = (x^2+y^2+z^2)^{-1/2}$

$\Delta\left((x^2+y^2+z^2)^{-1/2}\right)$

veya küresel koordinatlardaçözersiniz

$f(r;\theta,\phi) = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial f}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial f}{\partial\theta}\right)$ → 0

$+\frac{1}{r^2\sin^2\theta}\frac{\partial^2 f}{\partial\phi^2}$ → 0

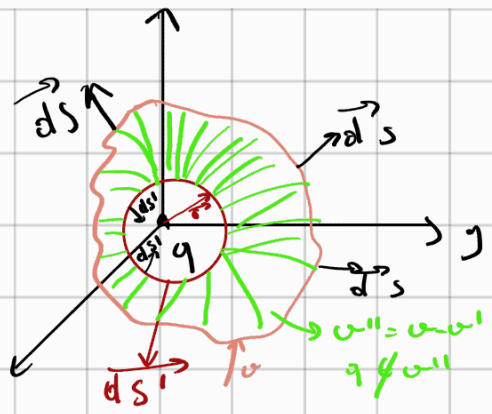
$f = \frac{1}{r} = f(r)$

$\frac{\partial(1/r)}{\partial\theta} = \frac{\partial(1/r)}{\partial\phi} = 0$

$\Delta(1/r) = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\underbrace{\frac{\partial}{\partial r}(1/r)}_{-1/r^2}\right) = \frac{1}{r^2}\frac{\partial}{\partial r}\left[-r^2\frac{1}{r^2}\right]$

$= -1\frac{\partial}{\partial r}(1) = 0 \Rightarrow (r\neq 0)$

$(r\neq 0)$ iseqq v ise $\int_S \vec{D}\cdot d\vec{s} = 0$



S' : r' yarıçaplı bir küre yüzeyi

ω'' Bölgesi $S' \cup S''$ yüzeylerini kapsar

$$\vec{r} \in \omega'' \Rightarrow \Delta(1/r) = 0$$

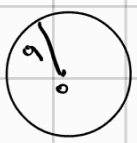
$$\Rightarrow \int_{\omega''} div\vec{D} \cdot d\omega'' = \int_{S'+S'} \vec{D}\left(\vec{ds} + \vec{ds'}\right)$$

$$\Rightarrow -\frac{q}{4\pi} \int_{\omega''} \overbrace{\Delta(1/r)}^{0} d\omega''$$

$$= \int_{S+S'} \vec{D}\left(\vec{ds} + \vec{ds'}\right) = 0 = \int_{S'} \vec{D}\,d\vec{S} + \int_{S'} \vec{D} \cdot \vec{ds'} = 0$$

$$\int_{S} \vec{D} \cdot \vec{dS} = ?? \quad (r = 0 \in \omega) \quad q \in \omega$$

$$\int_{S'} \vec{D} \cdot \vec{dS} = -\int_{S'} \vec{D} \cdot \vec{ds'}$$



$ds = a^2 \sin\theta\, d\theta\, d\phi$

$dv = r^2 \sin\theta\, d\theta\, d\phi$

q'nun S' de yarattığı $\vec{D}$: $\vec{D} = \frac{q}{4\pi} \frac{\vec{r'}}{(r')^3}$

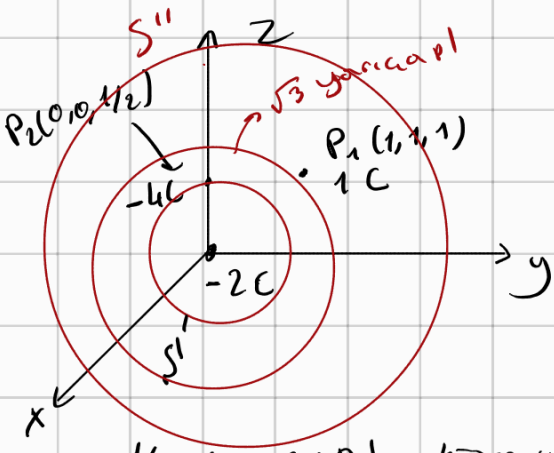
$$\vec{dS'} = -\vec{e}_{r'} \cdot \underbrace{r'^2 \sin\theta\, d\theta\, d\phi}_{dS'}$$

$$\vec{D} \cdot \vec{dS} = \frac{q}{4\pi} \frac{r'}{r'^3} \left(\vec{e}_{r'} \cdot r'^2 \sin\theta'\, d\theta'\, d\phi\right)$$

$$\oint_S \vec{D}\cdot d\vec{S}' = -\int_{\theta'=0}^{\pi}\int_{\phi'=0}^{2\pi} -\frac{q}{4\pi}\sin\theta'\,d\theta'\,d\phi'$$

$$= \frac{q}{4\pi}\underbrace{\int_{\theta'=0}^{\pi}\sin\theta'\,d\theta'}_{\to 2}\underbrace{\int_{\phi'=0}^{2\pi} d\phi'}_{\to 2\pi} = \frac{q}{4\pi}\cdot 4\pi = q \;//$$

$$\int_{S'} \vec{D}\,d\vec{S} = Q_{iç} + \frac{Q_{yüzey}}{2}$$



1/2 yarıçaplı küre yüzey

tekillik için koyulur yüzey üzerinde çünkü olduğu için

$$\oint_{S'} \vec{D}\cdot d\vec{S} = -2C + \left(\frac{-4C}{2}\right)$$

$$= -2C - 2C = -4C$$

$$[\vec{D}] = C/m^2$$

$$[\vec{E}] = V/m \qquad [V] = V_o$$

$$\oint_{S'} \vec{D}\cdot d\vec{S}' = -2C - 4C = -6C$$

$$\oint_{S''} \vec{D}\cdot d\vec{S}'' = -2C - 4C + 1C = -5C$$

Sınav 2 yada 3 soru

$$\int_S \vec{D}\cdot d\vec{S} = \int_v \operatorname{div}\vec{D}\cdot dv = \int_v \rho\, dv$$

(v'deki Q yükü)

$$= Q_{iç} //$$

$$\rho = \frac{dQ}{dv} \quad (v \text{ içindeki h.y.y. } [\rho] = C/m^3)$$

$$\boxed{\operatorname{div}\vec{D} = \rho} \rightarrow \text{maxwell}$$

Boşlukta

$$\vec{D} = \varepsilon_0 \vec{E} \Rightarrow \operatorname{div}\vec{E} = \frac{\rho}{\varepsilon_0}$$

$$\vec{E} = -\operatorname{grad} V$$

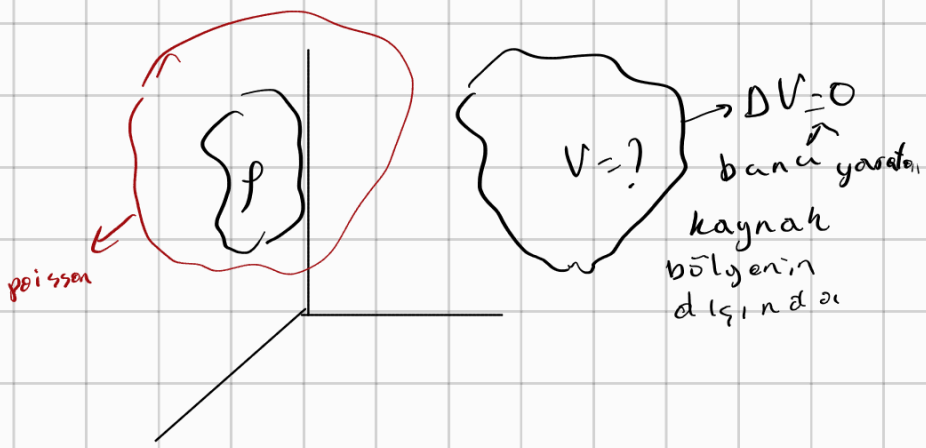
$$\rightarrow -\underbrace{\operatorname{div}\operatorname{grad} V}_{\Delta V} = \frac{\rho}{\varepsilon_0} \Rightarrow \Delta V = -\frac{\rho}{\varepsilon_0}$$

v'de $\rho \neq 0 \Rightarrow v$'deki V pot fonksiyonu

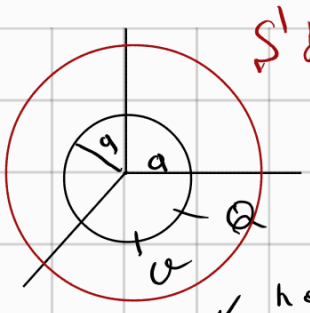
$$\Delta V = -\rho/\varepsilon_0 \rightarrow V = \frac{1}{4\pi\varepsilon_0}\int_v \frac{\rho}{r'}\, dv$$

↓ Poisson denklemi

v'de $\rho = 0$ (ya hiç yük yok) $\Rightarrow \boxed{\Delta V = 0}$ → Laplace



S.32
Pr 5



$v = \frac{4\pi a^3}{3}$

homojen yayılmış

$\rho = \frac{dQ}{dv} \Rightarrow \frac{Q}{v} = \rho = \frac{3Q}{4\pi a^3}$ (C/m^2)

$$Q = \int_v \rho \, dv = \int_{r=0}^{a} \int_{\vartheta=0}^{\pi} \int_{\phi=0}^{2\pi} \frac{3Q}{4\pi a^3} r^2 \sin\vartheta \, d\vartheta \, d\phi \, dr$$

$$= \frac{3Q}{4\pi a^3} \int_{r=0}^{a} r^2 dr \int_{\vartheta=0}^{\pi} \sin\vartheta \, d\vartheta \int_{\phi=0}^{2\pi} d\phi$$

$$= \frac{3Q}{4\pi a^3} \frac{a^3}{3} \cdot 2 \cdot 2\pi = Q$$

$V = V(r)$, $r > a$ $\quad \vec{E} = -\text{grad}\, V = \frac{-\partial V}{\partial r} \vec{e}_r$

$$\vec{D} = \varepsilon_0 \vec{E} = \underbrace{\varepsilon_0 \frac{\partial V}{\partial r}}_{D(r)} \vec{e}_r$$

$$d\vec{S} = r^2 \sin\alpha \, d\vartheta \, d\phi \, \vec{e}_r$$

$$\int_{S'_r} \vec{D} \cdot d\vec{S} = Q \Rightarrow \int_{S'_r} \underbrace{D(r)}_{\vec{D}} \underbrace{\vec{e}_\vartheta \cdot \vec{e}_r}_{1} \cdot r^2 \sin\vartheta \, d\vartheta \, d\phi = Q$$

$d\vec{S}_r$

$$\int_{\vartheta=0}^{\pi} \int_{\phi=0}^{2\pi} D(r) r^2 \sin\vartheta \, d\vartheta \, d\phi = Q$$

$$\Rightarrow D(r)\,r^2 \int_{\theta=0}^{\pi} \sin\theta\, d\theta \int_{\varphi=0}^{2\pi} d\varphi = Q$$

$$D(r)\, r^2\, 4\pi = Q$$

$$D(r) = \frac{Q}{4\pi r^2} \Rightarrow \vec{D} = \frac{Q}{4\pi r^2}\vec{e}_r$$

o zaman merkezde tek bir yük gibi küre davranır $r > a$ için

$$\vec{D}(r) = \frac{Q}{4\pi r^2}\vec{e}_r = -\varepsilon_0 \frac{\partial V}{\partial r}\vec{e}_r$$

$$V = \int \frac{\partial V}{\partial r} dr = \int \frac{-Q}{4\pi\varepsilon_0 r^2} dr \Rightarrow V(r) = \frac{Q}{4\pi\varepsilon_0}\frac{1}{r} + V_0$$

$$r \to \infty \Rightarrow V = 0 \Rightarrow V_0 = 0$$

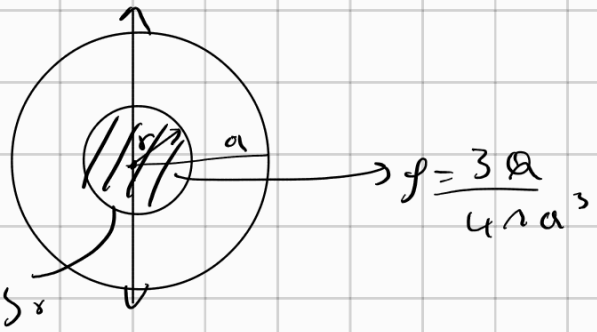
$$V(r) = \frac{Q}{4\pi\varepsilon_0}\frac{1}{r} + V_0$$

$$\lim_{r\to\infty} V(r) = V_0 = 0$$

$r > a$: $V = \frac{Q}{4\pi\varepsilon_0}\cdot\frac{1}{r}$

$\nabla V = \frac{Q}{4\pi\varepsilon_0}\nabla(1/r) = 0$

A $0 < r < a$



$$\vec{D} = D(r)\,\vec{e}_r$$

$$\int_{S_r} \vec{D}\cdot\vec{dS}_r \overset{\theta=0,\varphi=0}{\overset{\nearrow}{\nearrow}}{}^{2\pi}{}^{\pi} D(r)\,\underbrace{\vec{e}_r\cdot\vec{e}_r}_{1}\cdot r^2\sin\theta\,d\theta\,d\varphi = \int_{V_r}\rho\,dV_r$$

$$D(r)r^2\int_{\theta=0}^{\pi}\sin\theta\,d\theta\int_0^{2\pi}d\varphi = \int\frac{3Qa}{4\pi a^3}\,dV_r$$

$$\Rightarrow D(r)r^2\,4\pi = \frac{3Qa}{4\pi a^3}\,V_r = \frac{3Qa}{4\pi a^3}\,\frac{4\pi r^3}{3}$$

$$\int_{V_r} dV_r = V_r = \frac{4\pi r^3}{3}$$

$$\int_{V_r} dV_r = \int_0^r\int_{\theta=0}^{\pi}\int_{\varphi=0}^{2\pi} \underbrace{r^2\sin\theta\,d\theta\,d\varphi\,dr}_{dV_r = dr\,dS_r}$$

$$= \int_{r=0}^{r} r^2dr\,\underbrace{\int_{\vartheta=0}^{\pi}\sin\theta\,d\theta}_{2}\,\underbrace{\int_{\varphi=0}^{2\pi}d\varphi}_{2\pi}$$

$$\frac{r^3}{3}\Big|_0^r = \frac{r^3}{3}$$

$$\frac{4\pi r^3}{3}$$

$$\Rightarrow D(r)\,r^2\,4\pi = \frac{Q\cdot r^3}{a^3}$$

$$\Rightarrow D(r) = \frac{Qr}{4\pi a^3}$$

$$\vec{D} = \left(\frac{Q}{4\pi a^3}\,r\right)\vec{e}_r = \left(-\varepsilon_0\frac{\partial V}{\partial r}\right)\vec{e}_r$$

$$\frac{\partial V}{\partial r} = -\frac{Q}{4\pi\varepsilon_0 a^3} r$$

$$V = -\frac{Q}{8\pi\varepsilon_0 a^3} r^2 + \widetilde{V_0}$$

$$V|_{r=a^+} = V|_{r=a^-}$$

$$\frac{Q}{4\pi\varepsilon_0} \frac{1}{r}\Big|_{r=a} = -\frac{Q}{\pi\varepsilon_0 a^3} \cdot r^2\Big|_{r=a} + \widetilde{V_0}$$

$$\frac{Q}{4\pi\varepsilon_0} \frac{1}{a} = -\frac{Q}{8\pi\varepsilon_0 a} + \widetilde{V_0}$$

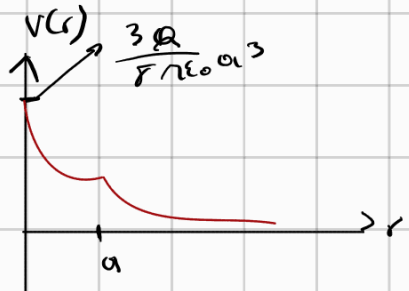
$$\widehat{V_0} = \frac{3Q}{8\pi\varepsilon_0 a}$$

$0 < r < a$

$$\boxed{V(r) = -\frac{Q}{8\pi\varepsilon_0 a} \left[\frac{r^2}{a^2} - 3\right]}$$

$r > a$:

$$V(r) = \frac{Q}{4\pi\varepsilon_0} \frac{1}{r}$$



$$V(r) = -\frac{Q r^2}{8\pi\varepsilon_0 a^2} + \frac{3Q}{8\pi\varepsilon_0 a^2}$$

$$\Delta V = \frac{1}{r^2} \frac{\partial}{\partial r}\left(r^2 \frac{\partial V}{\partial r}\right) = -\frac{Q}{8\pi\varepsilon_0 a^3} \Delta(r^2)$$

$$= -\frac{Q}{8\pi\varepsilon_0 a^3} \cdot \frac{1}{r^2} \cdot \frac{\partial}{\partial r}\left[r^2 \underbrace{\frac{\partial (r^2)}{\partial r}}_{2r^3} \right]$$

$$= -\frac{Q}{4\pi\varepsilon_0 a^3} \cdot \frac{3 \cancel{r^2}}{\cancel{r^2}} = -\frac{3Q}{4\pi a^3 \varepsilon_0}$$

$$= -\frac{\rho}{\varepsilon_0}$$